Chapter 8: Magnetostatic Forces

The following subjects will be discussed

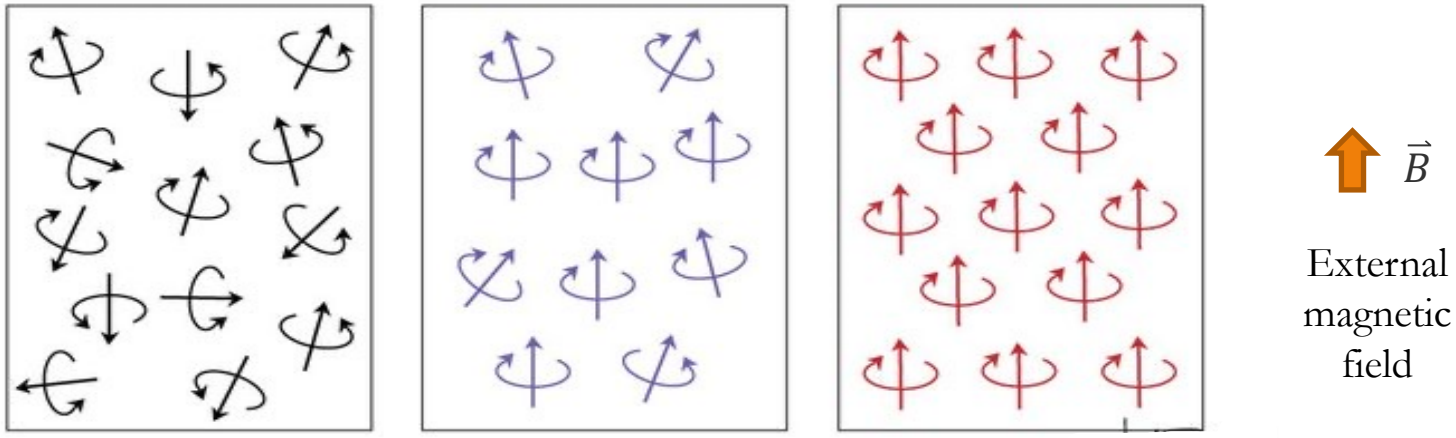
- Magnetization.
- Magnetostatic Boundary Conditions.
- Magnetic Energy Density.
- Definition of Inductance.

Magnetostatic Fields in Material

Materials can be classified based on the effect of external magnetostatic fields as magnetic and non-magnetic.

Magnetic materials are characterized by the ability of the internal random magnetic diploes to align parallel to applied external magnetic fields.

Non-magnetic materials have random magnetic dipoles that stay in random directions even with the presence of applied external magnetic fields.



Non-magnetic Material

Weak magnetic Material

Magnetic Material

Magnetization

When Magnetic materials are subjected to external magnetostatic fields, they become Magnetized.

This magnetization vector is proportional to the applied external field. $\vec{M} \propto \vec{H}$

$$\vec{M} = \gamma_m \vec{H}$$

γ_m : Magnetic susceptibility

Magnetic Flux Density inside a material is given by,

$$\vec{B} = \mu_o \left(\vec{H} + \vec{M}\right) \quad (\mathrm{Wb/m^2})$$

The magnetic flux inside the material is increased due to magnetization.

Magnetic Flux Density becomes,

$$\vec{B} = \mu_o \left(1 + \gamma_m\right) \vec{H} = \mu_o \mu_r \vec{H} = \mu \vec{H}$$

$\mu_r = 1 + \gamma_m$ is the relative permeability (unitless)

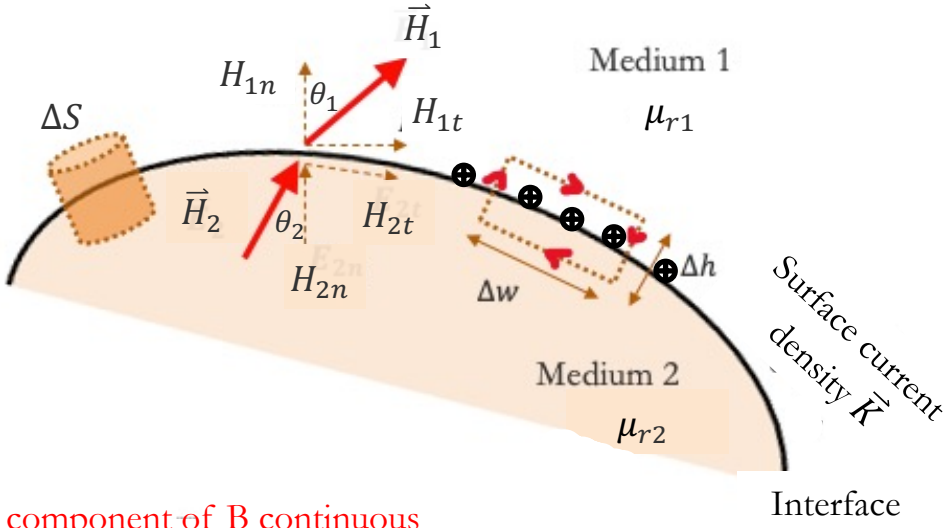
The the relative permeability is a property of the material.

Most materials in nature are non-magnetic ($\mu_r \approx 1$), even some good metals ($\mu_{r_Cu} \approx 0.999991$, $\mu_{r_Al} \approx 1.00002$). The class of materials that have high relative permeability is called ferromagnetics ($\mu_{r_Fe} \approx 5000$).

Magnetostatic Boundary Conditions

The relation between the magnetic field vector at the interface between two different media is given by the "boundary conditions".

To relate the normal components across the interface, evaluate the closed surface integral over some enclosure.



Normal component of B continuous

$$\oint_S \vec{B}\cdot\overrightarrow{dS} = 0 \;\rightarrow\; B_{1n}\Delta S - B_{2n}\Delta S = 0 \;\Rightarrow\; B_{1n} = B_{2n} \quad \text{and} \quad \mu_{r1}H_{1n} = \mu_{r2}H_{2n}$$

To relate the tangential components across the interface, evaluate Ampere's law around some contour.

$$\oint \vec{H}\cdot\overrightarrow{dl} = I_{encl} \;\rightarrow\; H_{1t}\Delta w + H_{1n}\frac{\Delta h}{2} + H_{2n}\frac{\Delta h}{2} - H_{2t}\Delta w - H_{1n}\frac{\Delta h}{2} - H_{2n}\frac{\Delta h}{2} = K\,\Delta w$$

The interface is defined as $\lim_{\Delta h\to 0}$ $\Rightarrow$ $\vec{H}_1\times\vec{a}_n - \vec{H}_2\times\vec{a}_n = \vec{K}$

For no surface current $H_{1t} = H_{2t}$ and $\dfrac{B_{1t}}{\mu_{r1}} = \dfrac{B_{2t}}{\mu_{r2}}$

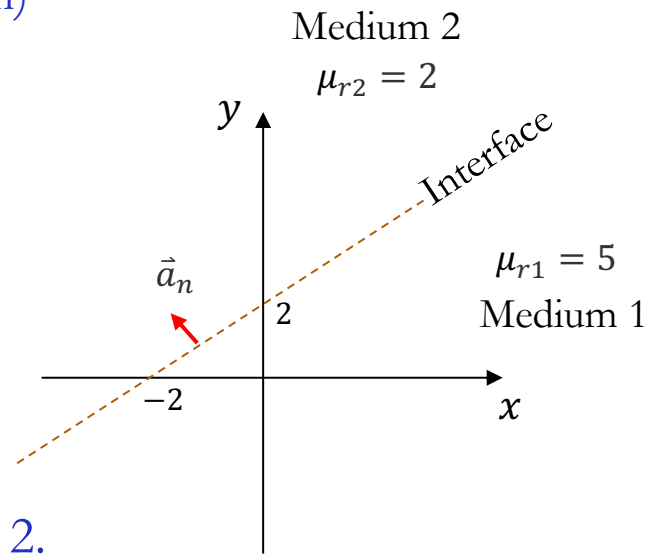
Tangential component of H continuous

Also $\dfrac{\tan\theta_1}{\tan\theta_2} = \dfrac{\mu_{r1}}{\mu_{r2}}$

Examples

(1) The plane $y - x - 2 = 0$ separates two magnetic materials. Given $\vec{K}$=0 and,

$$\vec{H}_1 = -2\,\vec{a}_x + 6\,\vec{a}_y + 4\,\vec{a}_z \quad \text{(A/m)}$$



Find the magnetization vector in medium 1.

$$\vec{M}_1 = \gamma_{m1}\vec{H}_1 = (5-1)(-2\,\vec{a}_x + 6\,\vec{a}_y + 4\,\vec{a}_z)$$

$$\vec{M}_1 = -8\,\vec{a}_x + 24\,\vec{a}_y + 16\,\vec{a}_z \quad \text{(A/m)}$$

Find the magnetic flux density in medium 1.

$$\vec{B}_1 = \mu_o\mu_r\vec{H}_1 = \mu_o(-10\,\vec{a}_x + 30\,\vec{a}_y + 20\,\vec{a}_z) \quad \text{(Wb/m}^2\text{)}$$

Find the magnetic field across the interface in medium 2.

$$\vec{a}_n = (-\vec{a}_x + \vec{a}_y)/\sqrt{2} \qquad \rightarrow \vec{H}_{1n} = (\vec{H}_1 \cdot \vec{a}_n)\,\vec{a}_n = -4\vec{a}_x + 4\vec{a}_y$$

$$\rightarrow \vec{H}_{1t} = \vec{H}_1 - \vec{H}_{1n} = 2\vec{a}_x + 2\vec{a}_y + 4\vec{a}_z$$

Apply B. C.

$$\vec{H}_{2t} = \vec{H}_{1t} = 2\vec{a}_x + 2\vec{a}_y + 4\vec{a}_z$$

$$\vec{H}_{2n} = \frac{\mu_{r1}}{\mu_{r2}}\vec{H}_{1n} = -10\vec{a}_x + 10\vec{a}_y$$

$$\vec{H}_2 = -8\,\vec{a}_x + 12\,\vec{a}_y + 4\,\vec{a}_z \quad \text{(A/m)}$$

Examples

(2) The plane $z = 0$ separates two regions. Region 1 ($z < 0$) is characterized by a relative magnetic permeability of 6, while that of region 2 ($z < 0$) is 4. The interface carries a constant current density given by, $\vec{K} = \left(\frac{1}{\mu_o}\right) \vec{a}_y$ (mA/m)

Given the flux density in region 2 as find the magnetic field in region 1.

$\vec{B}_2 = 5\,\vec{a}_x + 8\,\vec{a}_z$ (mWb/m^2)

The magnetic field in region 2 is

$\vec{H}_2 = \left(\frac{1}{4\mu_o}\right)(5\,\vec{a}_x + 8\,\vec{a}_z)$ (mA/m)

The normal is $\vec{a}_n = \vec{a}_z \rightarrow \vec{H}_{2t} = \left(\frac{5}{4\mu_o}\right)\vec{a}_x \qquad \vec{H}_{2n} = \left(\frac{2}{\mu_o}\right)\vec{a}_z$

Assuming that $\vec{H}_1 = H_{1x}\,\vec{a}_x + H_{1y}\,\vec{a}_y + H_{1z}\,\vec{a}_z \qquad \vec{B}_{1n} = \vec{B}_{2n} \rightarrow \vec{H}_{1n} = \left(\frac{8}{6\mu_o}\right)\vec{a}_z$

$$\vec{H}_1 \times \vec{a}_n - \vec{H}_2 \times \vec{a}_n = \vec{K} \rightarrow \left(H_{1x}\,\vec{a}_x + H_{1y}\,\vec{a}_y + \frac{8}{6\mu_o}\vec{a}_z\right) \times \vec{a}_z - \left(\frac{1}{4\mu_o}\right)(5\,\vec{a}_x + 8\,\vec{a}_z) \times \vec{a}_z = \left(\frac{1}{\mu_o}\right)\vec{a}_y$$

$$-H_x\,\vec{a}_y + H_y\,\vec{a}_x - \left(\frac{5}{4\mu_o}\right)H_z\,\vec{a}_y = \left(\frac{1}{\mu_o}\right)\vec{a}_y \rightarrow H_x = \frac{1}{4\mu_o} \qquad H_y = 0$$

$$\vec{H}_1 = \left(\frac{1}{\mu_o}\right)\left(\frac{1}{4}\,\vec{a}_x + \frac{8}{6}\,\vec{a}_z\right) \quad \text{(mA/m)}$$

Magnetic Energy Density

In the case of continuous charge distributions, defined by a charge density (ρ_L, ρ_s or ρ_v), the energy stored in the electric field is calculated using an integral equation.

$$W_m = \frac{1}{2}\int (\vec{B}\cdot\vec{H})\,dv = \frac{\mu}{2}\int |\vec{H}|^2 dv \quad \rightarrow \quad w_m = \frac{\varepsilon}{2}|\vec{H}|^2$$

Magnetostatic energy stored in the field (J)

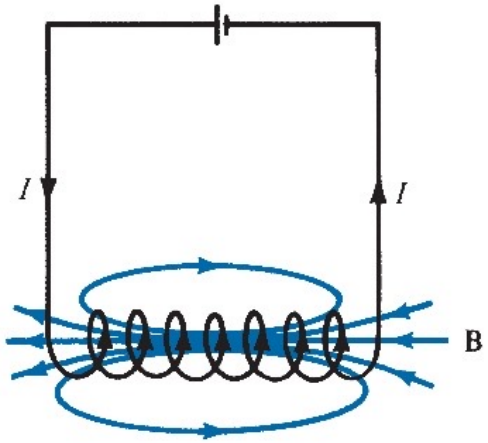
Magnetostatic energy density (J/m^3)

In the case electrostatic and magnetostatic fields exist in the same space, the total electromagnetic energy is given by.

$$W_{tot} = \frac{1}{2}\int \left(\varepsilon|\vec{E}|^2 + \mu|\vec{H}|^2\right) dv$$

Definition of Inductance

The amount of magnetic flux that can be generated for every Ampere of current in a given structure is called Inductance. It measured the ability of a structure to store magnetic energy as current passes through it.



$$L = \frac{N\psi}{I} = \frac{N\int(\vec{B}\cdot\overrightarrow{dA})}{I}$$

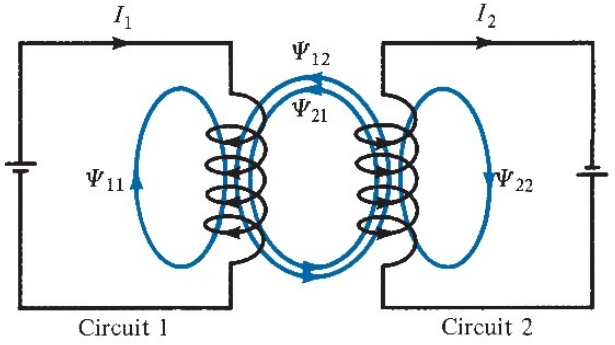
Unit is Henry (H)

L depends on

(1) Shape of structure
(2) dimensions
(3) filling material.

The ability of a circuit to store magnetic energy in another circuit is called the mutual inductance.

$$M_{21} = \frac{N_2\psi_2}{I_1} = \frac{N_2\int(\vec{B}_1\cdot\overrightarrow{dA_2})}{I_1}$$



Chapter 9: Maxwell's Equations

The following subjects will be discussed

- Faraday's Law of Induction.
- Maxwell's Equations.

Summary of Static Fields

Static EM Fields Equations

Differential Form	Integral Form
$\vec{\nabla}\cdot\vec{D}=\rho_v$	$\oint \vec{D}\cdot\overrightarrow{dS}=Q$
$\vec{\nabla}\times\vec{E}=0$	$\oint \vec{E}\cdot\overrightarrow{dl}=0$
$\vec{\nabla}\times\vec{H}=\vec{J}$	$\oint \vec{B}\cdot\overrightarrow{dA}=0$
$\vec{\nabla}\cdot\vec{B}=0$	$\oint \vec{H}\cdot\overrightarrow{dl}=I$

Continuity Equation

$$\vec{\nabla}\cdot\vec{J}+\frac{\partial}{\partial t}\rho_v=0$$

$$\vec{\nabla}\cdot\vec{J}=0 \qquad \text{DC case}$$

EM Boundary Conditions

$$E_{1t}=E_{2t} \qquad D_{1n}=D_{2n}+\rho_s$$

$$\vec{H}_1\times\vec{a}_n-\vec{H}_2\times\vec{a}_n=\vec{K} \qquad B_{1n}=B_{2n}$$

Constitutive Relations (material)

$$\vec{J}=\sigma\vec{E} \qquad \vec{B}=\mu\vec{H} \qquad \vec{D}=\varepsilon\vec{E}$$

EM Energy Density

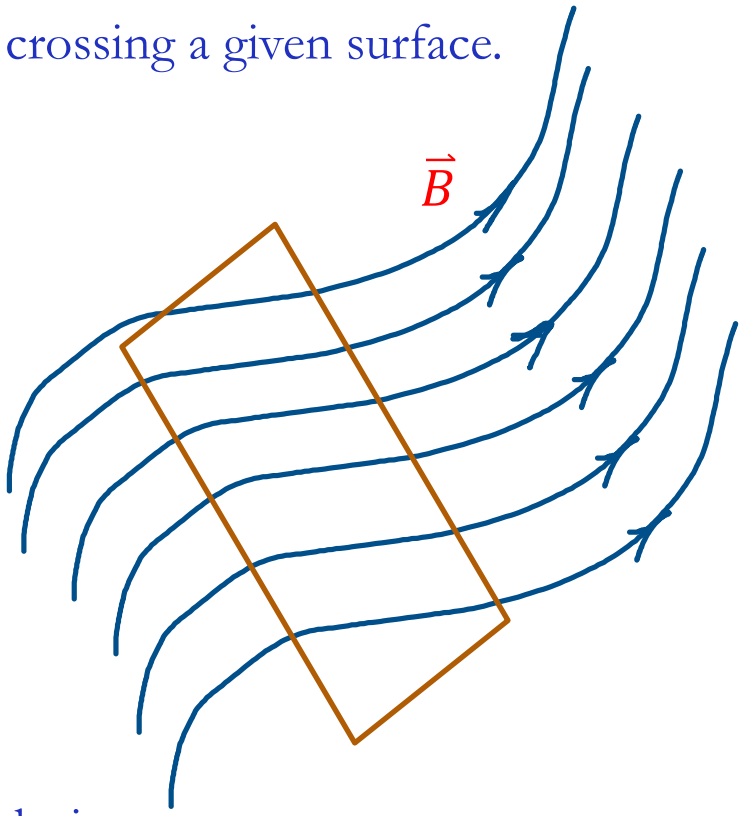
$$w_E=\frac{\varepsilon}{2}\left|\vec{E}\right|^2 \qquad w_m=\frac{\varepsilon}{2}\left|\vec{H}\right|^2$$

Faraday's Law

Magnetic flux is the number of magnetic field lines crossing a given surface.

Flux equals the surface integral of the field over the given surface.

$$\phi = \int \vec{B} \cdot \overrightarrow{dA}$$



If there is a change in the linkage in the wire (circuit) with respect to time, in any way, an emf will be generated within the wire (circuit)

$$Emf = -\frac{d\phi}{dt}$$

Faraday's Law Of induction

Ways to change linkage in time:

(1) B changing with time.

(2) Area changes with time

(3) Circuit is moving

(4) Combinations of the above.

Induced Electric Field

Let's find a direct relation between the change in magnetic field and the corresponding induced electric field.

For a time-varying magnetic field, we can write

$$Emf = -\frac{d\phi}{dt}$$

$$Emf = -\frac{d\phi}{dt} = -\int \frac{d\vec{B}}{dt} \cdot \overrightarrow{dA}$$

We know that

$$V = -\int \vec{E} \cdot \overrightarrow{dl} \quad \Rightarrow \quad Emf = \oint \vec{E} \cdot \overrightarrow{dl}$$

So, we have

$$\oint \vec{E} \cdot \overrightarrow{dl} = -\frac{d\phi}{dt} = -\int \frac{d\vec{B}}{dt} \cdot \overrightarrow{dA}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Note:
The electric field is generated even in the absence of a circuit!

Maxwell's Equations of Electromagnetics

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

$$\vec{\nabla} \cdot \vec{D} = \rho_v$$

Q & A

Send me an email if you have any questions.